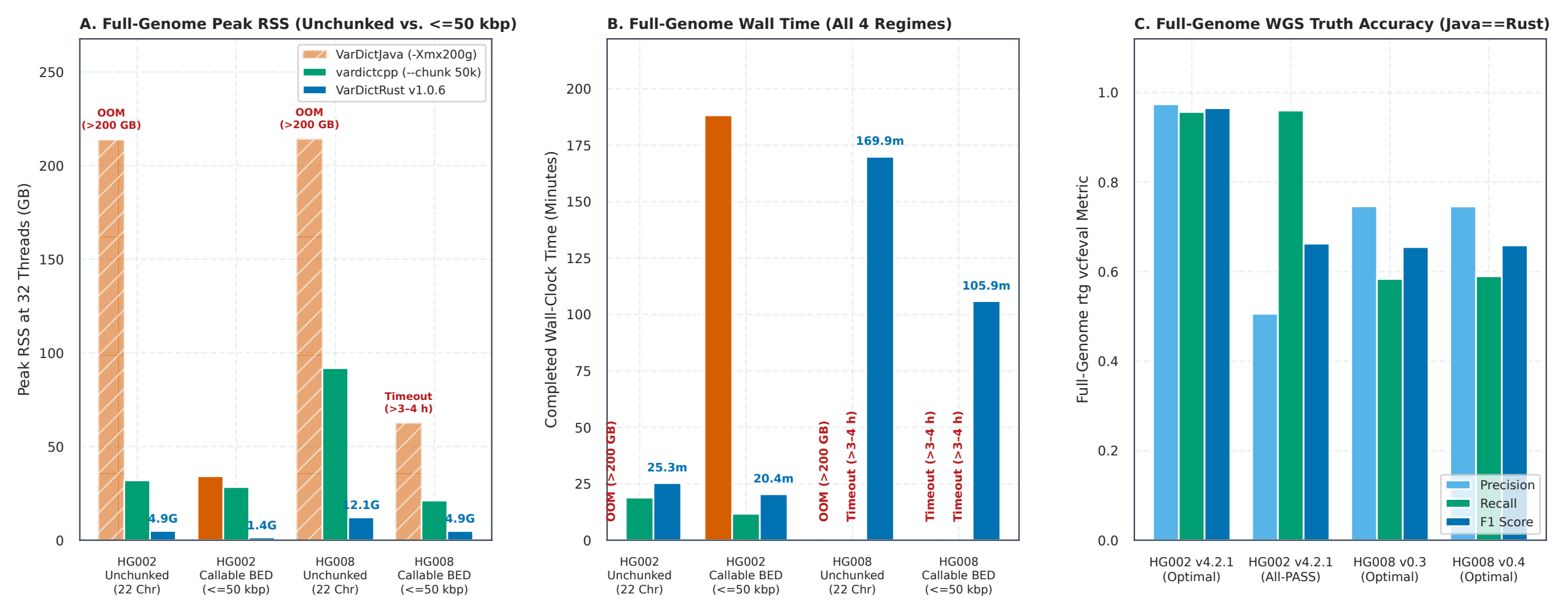
A. Full-Genome Peak RSS (Unchunked vs. <=50 kbp)
Peak RSS at 32 Threads (GB)
VarDictJava (-Xmx200g)
vardictcpp (--chunk 50k)
VarDictRust v1.0.6
OOM (>200 GB)
OOM (>200 GB)
Timeout (>3–4 h)
4.9G
1.4G
12.1G
4.9G
HG002 Unchunked (22 Chr)
HG002 Callable BED (<=50 kbp)
HG008 Unchunked (22 Chr)
HG008 Callable BED (<=50 kbp)
B. Full-Genome Wall Time (All 4 Regimes)
Completed Wall-Clock Time (Minutes)
OOM (>200 GB)
25.3m
20.4m
OOM (>200 GB)
Timeout (>3–4 h)
169.9m
Timeout (>3–4 h)
Timeout (>3–4 h)
105.9m
C. Full-Genome WGS Truth Accuracy (Java==Rust)
Full-Genome rtg vcfeval Metric
Precision
Recall
F1 Score
HG002 v4.2.1 (Optimal)
HG002 v4.2.1 (All-PASS)
HG008 v0.3 (Optimal)
HG008 v0.4 (Optimal)